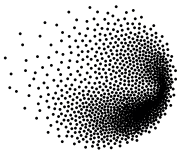




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and Sciences

Multi-Criteria Decision Analysis for the Evaluation of Nuclear Reactor Designs

TESI DI LAUREA MAGISTRALE IN
NUCLEAR ENGINEERING - INGEGNERIA NUCLEARE

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Abstract

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Parole chiave: qui, vanno, le parole chiave, della tesi

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Introduction

A brief History of Nuclear Power

Origins

The 1930s and 1940s witnessed a very fast development in nuclear science. Following James Chadwick's 1932 discovery of the neutron [13], Otto Hahn and Fritz Strassmann demonstrated nuclear fission in 1938 by producing barium through neutron bombardment of uranium [32]. This breakthrough enabled Enrico Fermi's team to construct the Chicago Pile-1 in 1942, the first artificial nuclear reactor capable of sustaining and controlling a chain reaction.

While these initial advancements were motivated by World War II and lead to the atomic bombings of Hiroshima (6 August 1945) and Nagasaki (9 August 1945), the scientific community recognized nuclear technology's peaceful potential. USA President Eisenhower's 1953 "Atoms for Peace" address to the United Nations [17] established the foundation for international nuclear cooperation, leading to the creation of both the International Atomic Energy Agency (IAEA) and the European Atomic Energy Community (EURATOM) in 1957. These institutions were tasked with promoting peaceful nuclear applications while serving as independent authorities for safety regulation and non-proliferation oversight.

The 1950s marked a period of rapid development of nuclear reactor designs, driven by dual civilian and military purposes. This era saw several key milestones in nuclear power generation: the Experimental Breeder Reactor I (EBR-I) in Idaho (USA) first produced electricity in 1951 (powering four light bulbs), followed by the Obninsk Nuclear Power Plant (USSR) in 1954, which became the first to feed approximately 5 MWe into a local grid. The world's first commercial-scale nuclear power station, Calder Hall-1 (UK), began operation in 1956, connecting to the national grid with an initial capacity of 50 MWe. The United States pioneered the light water reactor development (LWR), originally for naval propulsion purposes, launching the USS Nautilus in 1954. The first civilian application of the LWR technology in the USA was the Shippingport Atomic Power Station, which started operation in 1957 with a capacity of 60 MWe [38].

Parallel technological diversification occurred during this period. The United Kingdom developed gas-cooled Magnox reactors, optimized for both electricity production and plutonium generation, while the Soviet Union initiated development of the graphite-moderated, water-cooled RBMK design through experimental work at Obninsk. Canada's nuclear program focused on pressurized heavy water reactor (PHWR) technology, culminating in the 1962 Rolphoton demonstration reactor and the first commercial CANDU reactor at Douglas Point in 1968 [38].

Between the 60s and 70s a significant global expansion of nuclear power generation took place. While the Soviet Union developed their own version of the PWR technology, the VVER ("Water-Water Energetic Reactor"), numerous countries commissioned their first nuclear reactors: Argentina (Atucha-1, 1974), Armenia (Armenian-1, 1976), Belgium (BR-3, 1962), Bulgaria (1974), Italy (Latina, 1963), Japan (1963), Sweden (1964), Switzerland (1969), and Slovakia (1972) [38]. The 1973 oil crisis accelerated nuclear adoption as an energy security strategy. France implemented the Messmer Plan, an ambitious plan to expand the nuclear power plant fleet [27].

Two setbacks: 1986 and 2011

The nuclear expansion continued through the 1980s until the 1986 Chernobyl disaster, which constituted a critical inflection point for global nuclear development. The accident prompted widespread safety reviews and regulatory updates while public opposition intensified, particularly in Western Europe, leading several countries to reduce or halt their nuclear programs. Italy's 1987 referendum resulted in the complete phase-out of nuclear power, while other European nations experienced similar setbacks [48, 38].

By the 1990s, multiple countries had suspended nuclear deployment entirely, including Germany and Switzerland. France, previously the most active builder, connected only 10 reactors between 1990-1999, while the United States and United Kingdom deployed 4 and 1 reactors respectively during the same period. Italy completed its nuclear phase-out by 1990 with the permanent shutdown of all operating reactors. Following the post-Soviet economic transition, Russia resumed its nuclear program and has since continued expansion of its reactor fleet [38].

A nuclear renaissance emerged in the early 2000s, driven by concerns over climate change, energy security, and rising fossil fuel prices. This period saw the development of advanced reactor designs including the European Pressurized Reactor (EPR) and AP1000. However, the 2011 Fukushima Daiichi accident constituted another significant setback for nuclear expansion in Western nations. Projects faced substantial delays, cost overruns, and public

opposition, with France's Flamanville-3 EPR serving as a particularly illustrative case: experiencing budget increases to more than four times original estimates and construction timelines tripled from initial projections [51]. Germany accelerated its nuclear phase-out policy, while Italy reaffirmed its anti-nuclear position through a popular referendum [47].

Contrastingly, Eastern nations pursued an opposite trajectory. China, having initiated its nuclear program in the 1990s with Qinshan-1 (connected in 1991), has since maintained aggressive expansion. Over the past three decades, China has brought 56 additional reactors online and currently has 29 under construction, positioning itself to surpass both France and the United States in total reactor capacity within the coming decade [38].

Recent years

The current political scenario

In 2020, the COVID-19 pandemic exposed vulnerabilities in global economic systems and Russia's invasion of Ukraine in 2022 further revealed Europe's energy security dependencies [10, 76, 9]. Concurrently, decarbonization imperatives to combat climate change have prompted Western nations to reconsider nuclear energy's dual role: providing low-carbon, reliable baseload power while enhancing energy independence and system resilience [21].

This strategic reassessment of nuclear is evident in recent policy initiatives. The European Union's classification of nuclear power as a sustainable investment under its taxonomy for climate mitigation [22], alongside programs like the U.S. Department of Energy's Advanced Reactor Demonstration Program [82], are a clear indicator of the renewed interest in nuclear technologies.

This historical overview demonstrates that the initial decision to adopt nuclear power constitutes a strategic and political choice. However, the subsequent selection of a specific reactor design involves complex trade-offs across technical, economic, regulatory, social, environmental, and geopolitical aspects, particularly given the significant variations in national energy requirements.

The state of Nuclear Designs in 2025

As mentioned before, several technologies have been developed throughout the years, with some of them being split between opposing political sides, most notably the western implementation of the PWR technology and its soviet counterpart, the VVER. Nowadays the most common reactor type by far is the PWR, 306 out of the total 416 worldwide.

Followed by heavy water reactors (46) and BWRs (43). The current generation of these reactors are usually referred to as Generation III+ reactors, variations of previous iterations from the 90s (Gen III) updated to the latest safety standards (post 2011). Generation IV reactors are under development and employ innovative means of cooling, like molten salts or molten metals. SMRs on the other hand is a different approach to the building strategy of the reactor. For years nuclear power has seen bigger and bigger reactors, which in the latest year came with the problems of the so called mega-projects. While trying to pursue economies of scale at a certain point the trend is inverted catastrophically and massive over costs and delays afflicts these projects. SMRs want to solve this with smaller reactor, trying to standardize the design and produce components off site to improve on economies of multiples. Among SMRs designs there are both those based on novelty generation IV technologies and others that take advantage of the well known water-based technology

Challenges and solutions of our days

With a revived consideration for nuclear energy, a scrambled geopolitical theater and a lot of innovation, the task of committing to one design that policymakers might be faced with is definitely not easy. Modern day policies have to consider several factors, in the case of nuclear we have to keep in mind that commitments entail decade-long development, 60-year operational lifespans and substantial financial investments, at the same time we must not forget to consider grid infrastructure, public acceptance, and supply chain considerations. For this reason structured decision-support frameworks have been developed, like Multi-Criteria Decision Analysis (MCDA) which helps the policymakers to tackle decisions where many conflicting criterias have to be addressed.

The contemporary nuclear energy landscape presents policymakers with significant decision-making challenges. Renewed interest in nuclear power, coupled with geopolitical uncertainties and technological innovations, complicates the selection of an optimal reactor design. Modern energy policies must consider multiple intersecting factors: nuclear commitments involve decade-long development timelines, 60-year operational lifespans, and substantial financial investments, while simultaneously requiring evaluation of grid infrastructure compatibility, public dynamics, and supply chain vulnerabilities.

To address this complex decision-making environment, structured analytical frameworks have been developed, notably Multi-Criteria Decision Analysis (MCDA). This methodology provides systematic support for evaluating alternatives when multiple, often conflicting, criteria must be considered.

1 | Literature Review and Objectives

1.1. Literature Review

A review of relevant literature reveals numerous applications of MCDA methods across diverse fields. Focusing specifically on studies combining MCDA with nuclear power decision-making, several distinct approaches emerge:

Site selection studies represent the most common application, typically employing GIS-based software to evaluate potential locations for new nuclear installations within limited geographical regions. These analyses frequently incorporate quantitative non-technical factors such as population proximity [15] [7]. Abdullah et al. (2025) extend this approach by including social dimensions like public acceptance and tourism impacts, which present challenges for quantification [1]. Notably, Devanand et al. (2019) develop an algebraic optimization model to identify optimal SMR locations in Singapore, though like most site-focused studies, they favor a design without systematic comparison [16].

A second category compares different nuclear technologies. Kiser and Otero (2023) evaluates PWRs, SMRs, and Molten Salt Reactors [41], while Locatelli and Mancini (2011) compares large and small reactor designs [44]. Birikorang's study stands out for its regional focus on South Africa and comparison of three specific reactor models (NuScale SMR, EM2-HTGR, FTR-MSR), though its technological comparisons mix designs at different maturity levels [8]. Notably, all three studies utilize the Analytic Hierarchy Process (AHP) or its Fuzzy variant. The last one (Birikorang) draws on the IAEA's INPRO methodology for advanced nuclear technology comparison [36].

Broader energy comparison studies by Hernández-Torres (2025), Pasaoglu (2018), and Atilgan (2016) evaluate multiple generation options (typically eight alternatives) employing AHP-TOPSIS, AHP, and MAVT methodologies respectively. These studies use generic, averaged parameters for each technology class [33, 65, 5]. In contrast, Castellazo (2014) adopts a more precise approach using Multi-Attribute Value Theory (MAVT)

to compare thirteen energy technologies, employing specific reference plants (e.g. EPR reactor for nuclear power) rather than generalized technology averages [72].

Three studies show particular relevance to the current research. Zolkaffly et al. (2014) evaluates nuclear plant models in Malaysia but suffers from outdated data and neglects non-technical aspects [92]. Topaloğlu (2025) applies the Analytic Network Process (ANP) to compare reactor types (PWR, BWR, PHWR) and locations, though social considerations remain limited to site-specific factors [80]. Goushis (2025) employs TOPSIS to assess SMR deployment in the EU for coal replacement, but focuses on a single reactor model rather than comparative technology evaluation [31].

The literature review reveals three gaps in nuclear energy decision-making research. First, most studies focus solely on site selection rather than comparing different nuclear technologies. Second, comparisons between nuclear designs lack consideration of a specific decision-making context. Third, studies that do consider specific decision-making context, examine various energy generation options as broad technology categories (e.g., nuclear, solar, wind) rather than conducting detailed analyses of specific reactor models or power plant designs.

This study addresses these research gaps by applying Multi-Criteria Decision Analysis (MCDA), specifically employing Multi-Attribute Value Theory (MAVT), to compare reactor designs. The analysis explicitly incorporates country-specific conditions and examines their influence on the ranking of different nuclear installations.

1.2. Applying MAVT to choose a Reactor

This study evaluates a selected group of nuclear reactor designs to determine their suitability for different national contexts. By comparing four European countries with distinct energy profiles, the analysis will also explore how factors such as demand patterns, grid flexibility, and policy priorities, shape the ranking of reactor designs.

The methodology relies on Multi-Criteria Decision Analysis (MCDA) to systematically evaluate reactor designs by taking into account real world data, estimates, and expert judgment. Specifically, this study uses Multi-Attribute Value Theory (MAVT) which, first presented by Keeney in 1996 [40], transforms qualitative and quantitative data into comparable metrics (values) by means of value functions, then aggregates values by hierarchical weighting to produce a composite score for each option.

Here i will put the signposts to do the overview of the thesis structure... once i wrote it all... In the following section we are going to discuss the selection of reactor models then.... etc etc etc

2 | Preparation

2.1. Criteria for Reactor Selection in Comparative Analysis

This study focuses on a curated set of reactor designs at an advanced stage of development. The selection process begins with the the IAEA "Nuclear Reactors in the World" [56], which catalogs operational, under-construction, and decommissioned reactors globally.

The initial selection process applied two primary exclusion criteria. First, designs were eliminated based on supplier origin, removing all reactors associated with countries unlikely to engage in nuclear technology transfer with Europe under the current geopolitical climate. This filter specifically excluded reactors developed in or supplied by Russia, China, and other nations maintaining restricted technology-sharing agreements with Western markets.

Second, reactors commissioned before 2000 were excluded from consideration, as their designs represent outdated technological standards. The complete filtering methodology is documented in the accompanying analysis notebook (PRIS Data Analysis.ipynb), which details how the initial dataset was reduced to nine distinct reactor models.

From this pre-selected group, three designs - the EPR-1750, APR-1400, and AP-1000 - were prioritized for comprehensive analysis. It should be noted that while the EPR (1650 MWe) and EPR-1750 share significant design similarities, they are treated as separate models in this study. Additionally, the Advanced Boiling Water Reactor (ABWR), though listed as operational in the IAEA ARIS database, was excluded following verification through the PRIS database, which indicates all ABWR units have suspended operation since 2011 [38]. The selection process and its outcomes are summarized in Table 2.1.

Subsequently, the IAEA Advanced Reactors Information System (ARIS) database [2], was consulted to identify advanced reactor models classified as under design, under regulatory

review, under construction, or operational. Demonstrations units and prototypes were excluded from this set, yielding 10 candidate designs. To avoid redundancy, models already included in the prior selection (e.g., Generation III+ designs captured in both datasets) were removed. Each remaining design was then assessed for development status, technological readiness, and applicability to European energy systems as summarized in Table 2.2.

The overall selection produced a set of six reactors, three Generation III+ and three SMRs. Their main characteristics are summarized in Table 2.3. The most notable differences are the integrated designs of the BWRX-300 and Nuscale SMR which eliminate the external primary loop in favour of steam generator inside the reactor pressure vessel. This allows for compact construction and allows to exploit natural circulation for cooling. Then the Nuscale SMR is the only model designed to be deployed in a multi unit configuration, with up to 12 modules in the same containment building. **Maybe here add a paragraph for each design to explain their main features.**

Reactor Name	Country	Technology-sharing	Commercial Operation	Up Date	to Decision
APR1400	S.Korea	✓	✓	✓	Selected
CAREM Prototyp	Argentina	✓	✗	✓	Rejected
PRE KONVOI	Germany	✓	✓	✗	Rejected
EPR 1750	France	✓	✓	✓	Selected
AP1000	USA	✓	✓	✓	Selected
ABWR	Japan	✓	✗	✓	Rejected
APWR	Japan	✓	✗	✓	Rejected
EPR	France	✓	✓	✗	Rejected
OPR-1000	S.Korea	✓	✓	✗	Rejected

Table 2.1: Selection of reactor models for analysis

Reactor Name	Country	Used	Reasoning
APR+	S.Korea	✗	As reported in the manufacturer website [20] this design is mean to be an "indigenous reactor cleared of obstacles to export"
ATMEA	Japan, France	✗	Joint venture was abandoned in 2019 [6]
BWRX300	USA, Canada	✓	Design from GE-Hitachi, a lot of support from US and Canada as well as interes in the EU [53] [54].
i-SMR	S.Korea	✗	Never expressed interest in development outside south Asia and the middle east, no updates since early 2024.
iMSR400	US	✗	No interest overseas, project seems to be at an early stage of development.
Natrium	US	✗	Demo plant [79].
Nuward	France	✗	After the change in direction of 2024 the project has been set back [6].
Nuscale	US	✓	Even if there was some financial controversy in 2024 [6]. the project seems to be still moving forward in terms of goverment founding [6] and regulatory review [6].
PRISM	US	✗	GE-Hitachi considers it a "concept ... currently being put into practice in two reactors: Natrium and ARC-100" [6].
Rolls Royce SMR	UK	✓	It's moving forward and from the technological point of view this is the simples as it it litteraly a scaled down PWR[6].
SMART	S.Korea, Saudi Arabia	✗	Plan to export only to south-east asia and middle east [55].

Table 2.2: Selection of advanced reactor models for analysis

Reactor Name	Type	Construction	P_e (MWe)	Supplier
APR1400	PWR	1 Unit	1340	KEPCO (S.Korea)
EPR 1750	PWR	1 Unit	1630	Framatome (France)
AP1000	PWR	1 Unit	1120	Westinghouse (USA)
BWRX300	i-BWR	1 Unit	300	GE-Hitachi (USA-Canada)
Rolls Royce SMR	PWR	4 to 12 Units	470	Rolls Royce (UK)
Nuscale	i-PWR	1 Unit	77	Nuscale Power (USA)

Table 2.3: Final selection of reactor models for analysis

2.2. Country Selection for Comparative Analysis

This study evaluates four European countries: France, Italy, Poland and Switzerland, to assess how national energy contexts influence the ranking of nuclear reactor designs.

In Italy, the energy market is divided into different zones, each with its own demand patterns. For this analysis, Northern Italy serves as the focal region being the country's most industrialized and energy-intensive area[6]. It is characterized by exceptionally high electricity demand, a strong reliance on imports, and above-average greenhouse gas emission intensity, making it a critical bottleneck in Italy's decarbonization efforts. Additionally, the country's historical political volatility introduces uncertainty into long-term energy planning.

France, with $\sim 70\%$ of its electricity generated by nuclear power[6], represents a mature nuclear market. Here, the marginal gains from additional reactors are likely minimal, offering insight into how reactor rankings shift in a saturated nuclear energy landscape.

Poland, heavily dependent on coal and among Europe's highest GHG emitters [6], presents an opposite case. As the country seeks to modernize its grid and reduce emissions through nuclear energy [6], it provides an opportunity to assess which reactor designs could best support its transition while meeting growing energy demand.

Finally, Switzerland provides a unique case due to its low electricity demand, hydropower-dominated grid, and prudent nuclear policy. Its constrained grid capacity and emphasis on energy independence make it ideal for assessing small modular reactors (SMRs) and their role in complementing renewables.

2.3. Attribute Definition

Throughout this section we will define the indicators.

2.3.1. Technical parameters

...

2.3.2. MOX Fuel Compatibility

Mixed Oxide (MOX) fuel is produced through spent nuclear fuel reprocessing and offers a solution for partially closing the nuclear fuel cycle in thermal reactors. While this process generates some secondary waste, it provides several strategic advantages: reducing final nuclear waste volumes, minimizing mining waste, and decreasing dependence on uranium imports. The latter is particularly relevant for Europe, which lacks significant domestic uranium deposits. However, reprocessing also presents proliferation risks since the extracted plutonium could potentially be diverted for weapons production. Due to these competing factors, nuclear fuel reprocessing is a fundamental strategic decision; the United States has foregone this practice, while France and UK have implemented the process in their nuclear programs [14].

Given the European context of this study, MOX compatibility is considered a relevant reactor attribute. The evaluation methodology assigned each reactor design an initial score (0-10) based on its maximum MOX core fraction capability, followed by qualitative adjustments reflecting design-specific features. The complete scoring process and results are detailed in Table 2.4.

Reactor Model	MOX[%]	Considerations	Score	Sources
APR1400	33	-	3/10	[2]
EPR 1750	100	50% without modifications, to reach 100% "design adaptations which are not significantly costly" are required.	9/10	[2, 4]
AP1000	50	50% without modification, to reach 100% major modifications are required	6/10	[2, 25]
BWRX-300	0	GNF-2 Not manufactured with MOX	0/10	[11]
Rolls Royce SMR	0	Rolls Royce states that they are not planning to use MOX	0/10	[68]
NuScale	0	Not licensing for MOX but a study reveals support is possible without design modifications	1/10	[87, 59]

Table 2.4: MOX fuel compatibility scores and maximum MOX fraction

2.3.3. Technological Readiness

Expert consultation

2.3.4. Safety

To be discussed

2.3.5. Capital Costs

DRAFT

Option	PROs	CONs
Use NOAK estimates	Use of estimates in all cases, comparable values	It's not realistic and not fair towards more mature designs
Use FOAK estimates/data	Better data for three large reactors	We are mixing up real data and estimates
Use NOAK and apply FOAK-to-NOAK conversion	Comparable values, with awareness of current know-how	rely on a model
Present NOAK and FOAK to experts along with technological readiness overview and let them evaluate	As before	rely on expert judgment

Table 2.5: Options overview

The same applies to commissioning time.

It is important to note that these costs refer to an Nth of a kind installation and will not be representative of the FOAK costs. Moreover as detailed by Rothwell 2022 [71] the costs of recent nuclear installations in Europe and the US have been significantly higher than those announced at the start of the project. For this reason we will apply a FOAK-to-NOAK factor to better represent the real costs. For example the EPR-1600 announced costs for Flamanville-3 was $1886 \text{ USD}_{2020}/kW_e$ but have risen to $8600 \text{ USD}_{2020}/kW_e$ [52]. All Values are summarized in Table 2.6.

For reference, the capital costs of the same plants as the previous section have been

reported by the US Energy Information Administration [83].

We want to clarify that the data has all been converted to 2020 USD since the main source of data [75] and [52] reports all costs in this currency and year.

Reactor Model	Capital Cost of NOAK [\$/kWe]	Capital Cost of FOAK [\$/kWe]	Data Source
APR-1400	1828	2410	[75, 52]
EPR-1750	2150	8600	[75, 52]
AP-1000	2000	8600	[75, 52]
BWRX-300	2318	3467	[67]
Rolls Royce SMR	6050	7408	[66]
NuScale	2850	3600	[60]
Other Power Plants			
Onshore Wind	-	1265	[83]
Solar PV + Battery	-	2175	[83]
Hydroelectric	-	6007	[83]

Table 2.6: Capital costs for selected reactor models

2.3.6. Commissioning Time

DRAFT

See observations in the previous section

I would apply a +0.5 years to planned commissioning time on the APR and AP based on the planned commissioning time of the EPR that differs by 0.5 between UK and Chinese plannings.

Reactor Model	Planned Commissioning Time	Commis- Last unit Commis- sioning Time	Data Source
APR1400	5.0 in Korea	10 and 8 in Korea	[51]
EPR 1750	5.0 in Europe, 4.5 in China	9 in Taishan, 17 in Flamanville and Olkiluoto	[51]
AP1000	5.0 in USA	9 in Sanmen, 10 and 11 at Vogtle	[51]
BWRX-300	2 ~ 3	NA	[29]
Rolls Royce SMR	4.0	NA	[69]
NuScale	3.0	NA	[61]

Table 2.7: Technological readiness and planned commissioning time

2.3.7. Operational Costs

Lascks presentation of the attribute

Steigerwald et al. (2023) [75] provides comprehensive cost analyses for future nuclear installations, including capital and operational expenditures for both SMRs and large reactors. The study reports operational costs in USD_{2020}/MWh_e for EPR and APR1400 designs, as well as a "Long-Term US" category which this analysis uses as AP1000 costs. For SMRs, operational costs were originally presented in power-based units (USD_{2020}/MW_e), requiring conversion to energy-based measurements (USD_{2020}/MWh_e) using a Gaussian distribution of capacity factors ($\mu = 90\%$, $\sigma = 5\%$) according to:

$$\frac{USD_{2020}}{MWh_e} = \frac{USD_{2020}}{MW_e} \cdot \frac{1}{CF \cdot 8760 \frac{h}{year}}$$

The same study reports fuel costs of $6.23USD_{2020}/MWh_e$ for PWR-based technologies with the exceptions of NuScale at $7.15USD_{2020}/MWh_e$ and the only BWR reactor at $29.55USD_{2020}/MWh_e$. For comparative context, operational cost estimates for utility-scale onshore wind, hydroelectric, and solar PV (with single axis tracking and DC battery storage) were obtained from the U.S. Energy Information Administration [83]. All cost data have been standardized to 2020 USD to maintain consistency with the primary data sources [75, 52], values are presented in Table 2.8.

Reactor Model	Operative Cost [\$/MWh]	Data Source
APR-1400	18.44	[75]
EPR-1750	14.26	[75]
AP-1000	18.69	[75]
BWRX-300	18.30 ± 0.82	[75]
Rolls Royce SMR	65.68 ± 2.94	[75]
NuScale	22.76 ± 1.02	[75]
Other Power Plants		
Onshore Wind	28.08	[83]
Solar PV + Battery	33.33	[83]
Hydroelectric	28.49	[83]

Table 2.8: Economic costs for selected reactor models

2.3.8. Potential Impact on the energy market

To write

2.3.9. Potential Impact on greenhouse gas emissions

To write

2.3.10. Waste

To write

2.3.11. Size - Nominal output power

To write

2.3.12. Load Following Capabilities

Load-following capability can be quantitatively assessed using characteristic curves that represent ramp-up and ramp-down rates as a function of load (expressed as percentage of nominal power $\%P_{nom}$), as illustrated in Figure 2.1. To facilitate comparative analysis, these performance curves are reduced to scalar values through integration of the area under said curve.

For benchmarking purposes, load-following capabilities were calculated for two modern gas turbine configurations: A heavy-duty combined cycle plant featuring two GE Vernova 9HA.02 gas turbines (2015 data) [30] and an equivalent plant configuration utilizing aeroderivative GE LM2500XPRESS+G5 DLE UPT gas turbines (2025 data) [28]. Hydroelectric plants are also included as a reference for the highest flexibility, a 1989 publication by the US Office of Technology Assessment [81] reports that the response rate of large ($>60\text{MW}$) hydroelectric plants ranges from 4 to 6 $\%P_{nom}/\text{s}$, which translates to 240 to 360 $\%P_{nom}/\text{min}$. A graph from a 1994 conference reported in a technical analysis from 2009 shows that some hydro plant can achieve 0 to 100% in much less than a minute and from black startup to full power in around 5 minutes. For the comparison we indicate greater than 100%. [73, 49]

To establish an upper benchmark hydroelectric plants were included in the comparative analysis. A 1989 U.S. Office of Technology Assessment report [81] documents that large ($>60\text{ MW}$) hydroelectric facilities demonstrate remarkable load-following capabilities, with response rates ranging from 4 to 6 $\%P_{nom}/\text{s}$ (equivalent to 240-360 $\%P_{nom}/\text{min}$). Additional performance data from a 1994 conference presentation, cited in a 2009 technical analysis [73, 49], indicates that certain hydroelectric installations can achieve 0-100% load changes in under one minute and black start to full power operation in approximately five

minutes. For the purposes of this comparison, hydroelectric flexibility is conservatively represented as $>100\%P_{nom}/\text{min}$.

The results of these analyses are presented in Table 2.9.

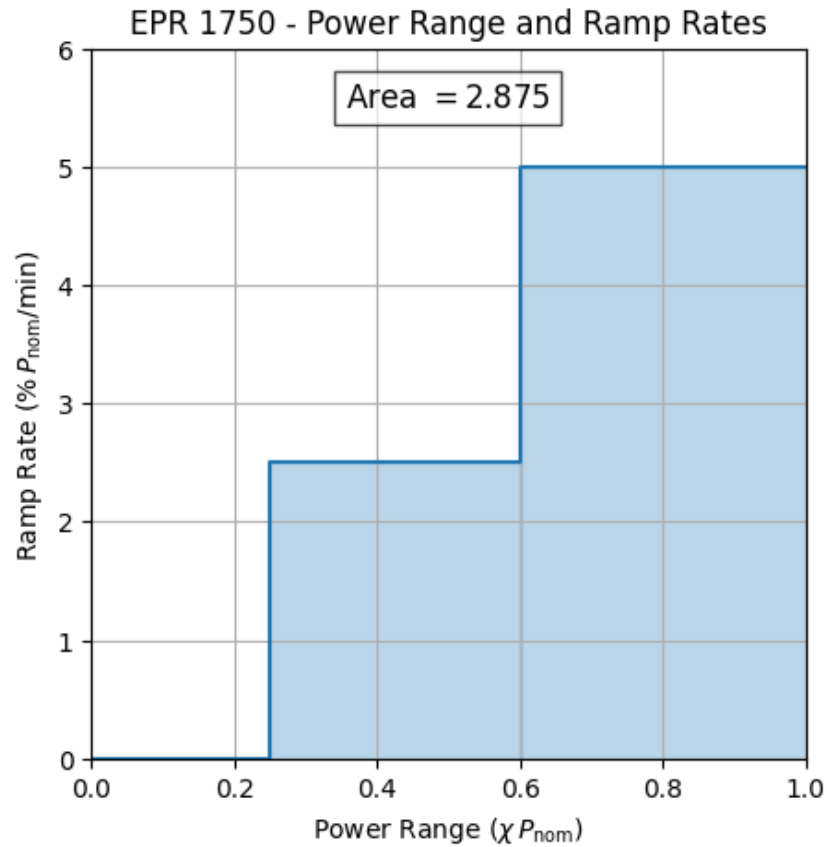


Figure 2.1: Example load following capability curve for EPR-1750

Reactor Model	Load Following Capability $\left[\frac{\%P_{nom}^2}{min} \right]$	Data Source
APR1400	0.21	[2]
EPR 1750	2.88	[62]
AP1000	4.25	[2]
BWRX-300	0.25	[2]
Rolls Royce SMR	2.00	[2]
NuScale	2.00	[19]
GE 9HA.02 2xCC	7.02	[30]
GE LM2500XPRESS+G5 2xCC	40.54	[28]
Hydroelectric	>100	[81, 73, 49]

Table 2.9: Load following capabilities and power output

3 | Analysis

4 | Results

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A | Appendix A

If you need to include an appendix to support the research in your thesis, you can place it at the end of the manuscript. An appendix contains supplementary material (figures, tables, data, codes, mathematical proofs, surveys, . . .) which supplement the main results contained in the previous chapters.

B | Appendix B

It may be necessary to include another appendix to better organize the presentation of supplementary material.

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List of Symbols

Variable	Description	SI unit
$\boldsymbol{u}$	solid displacement	m
$\boldsymbol{u}_f$	fluid displacement	m

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